

ECB1323 ANTENNA AND MILLIMETRE WAVE COMMUNICATION

BANDWIDTH ENHANCEMENT OF A PRINTED INVERTED-F ANTENNA AT 5 GHZ

A PROJECT REPORT

Submitted by

MOHAMED SALMAN AL FARISH A (927623BEC135)

MOHAMMED AATHIL J (927623BEC136)

MUKESH M (927623BEC141)

in partial fulfilment for the award of the degree

of

BACHELOR OF ENGINEERING

in

ELECTRONICS AND COMMUNICATION ENGINEERING

M.KUMARASAMY COLLEGE OF ENGINEERING, KARUR

MAY 2026

M.KUMARASAMY COLLEGE OF ENGINEERING, KARUR

(Autonomous Institution affiliated to Anna University, Chennai)

BONAFIDE CERTIFICATE

Certified that this project report “**BANDWIDTH ENHANCEMENT OF A PRINTED INVERTED-F ANTENNA AT 5 GHZ**” is the bonafide work of “**MOHAMED SALMAN AL FARISH A (927623BEC135), MOHAMMED AATHIL J (927623BEC136), MUKESH M (927623BEC141)**” who carried out the project work during the academic year 2025- 2026 under my supervision.

SIGNATURE

Dr.N.Mahendran, M.E., Ph.D

HEAD OF THE DEPARTMENT

Professor,

Department of Electronics and
Communication Engineering,

M.Kumarasamy College of Engineering,
Thalavapalayam, Karur-639113

SIGNATURE

Mr. P.Arun Kumar, M.Tech

SUPERVISOR

Associate Professor,

Department of Electronics and
Communication Engineering,

M.Kumarasamy College of Engineering,
Thalavapalayam, Karur-639113.

This Project Work ECB1323 ANTENNA AND MILLIMETER WAVE COMMUNICATION report has been submitted for the End Semester Project viva voce Examination held on _____.

INTERNAL EXAMINER

VISION AND MISSION OF THE INSTITUTION

VISION

To emerge as a leader among the top institutions in the field of technical education.

MISSION

- Produce smart technocrats with empirical knowledge who can surmount the global challenges.
- Create a diverse, fully-engaged learner-centric campus environment to provide quality education to the students.
- Maintain mutually beneficial partnerships with our alumni, industry and professional associations.

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

VISION

To empower the Electronics and Communication Engineering students with emerging technologies, professionalism, innovative research and social responsibility.

MISSION

- M1: Attain the academic excellence through innovative teaching learning process, research areas & laboratories and Consultancy projects.
- M2: Inculcate the students in problem solving and lifelong learning ability.
- M3: Provide entrepreneurial skills and leadership qualities.
- M4: Render the technical knowledge and skills of faculty members.

PROGRAMME EDUCATIONAL OBJECTIVES (PEOs)

PEO1: Core Competence: Graduates will have a successful career in academia or industry associated with Electronics and Communication Engineering.

PEO2: Professionalism: Graduates will provide feasible solutions for the challenging problems through comprehensive research and innovation in the allied areas of Electronics and Communication Engineering.

PEO3: Lifelong Learning: Graduates will contribute to the social needs through lifelong learning, practicing professional ethics and leadership quality.

PROGRAM OUTCOMES(POs)

PO1: Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems to reach substantiated conclusions considering sustainable development.

PO3: Design/development of solutions: Design innovative solutions for complex engineering problems and develop systems/components considering public health, safety, society, environment, and sustainability.

PO4: Investigation of Complex Problems: Conduct investigations using research methods including experiment design, modelling, analysis, and interpretation of data to provide valid conclusions.

PO5: Engineering Tool Usage: Create, select, and apply modern engineering and IT tools, techniques, and resources including modelling and simulation while recognizing their limitations.

PO6: The Engineer and the World: Analyze societal, environmental, economic, health, safety, legal, and cultural impacts of engineering solutions for sustainable development.

PO7: Ethics: Apply ethical principles, professional responsibilities, human values, diversity, and inclusion while adhering to national and international laws.

PO8: Individual and Collaborative Team Work: Function effectively as an individual and as a member or leader in multidisciplinary and diverse teams.

PO9: Communication: Communicate effectively with the engineering community and society through reports, presentations, and technical documentation.

PO10: Project Management and Finance: Apply engineering management principles and economic decision-making in project planning and multidisciplinary environments.

PO11: Life-long learning: Recognize the need for continuous learning and adapt to new and emerging technologies throughout professional life.

PROGRAM SPECIFIC OUTCOMES (PSOs)

PSO1: Applying knowledge in various areas, like Electronics, Communications, Signal processing, VLSI, Embedded systems etc., in the design and implementation of Engineering application.

PSO2: Able to solve complex problems in Electronics and Communication Engineering with analytical and managerial skills either independently or in team using latest hardware and software tools to fulfil the industrial expectations.

ACKNOWLEDGEMENT

We gratefully remember our beloved **Founder Chairman, (Late) Thiru. M. Kumarasamy**, whose vision and legacy laid the foundation for our education and inspired us to successfully complete this project.

We extend our sincere thanks to **Dr.K.Ramakrishnan, Chairman**, and **Mr.K.R.Charun Kumar, Joint Secretary**, for providing excellent infrastructure and continuous support throughout our academic journey.

We are privileged to extend our heartfelt thanks to our respected Principal, **Dr. B. S. Murugan, B.Tech., M.Tech., Ph.D.**, for providing us with a conducive environment and constant encouragement to pursue this project work.

We sincerely thank **Dr. N. Mahendran, B.E., M.E., Ph.D., Professor and Head, Department of Electronics and Communication Engineering**, for his continuous support, valuable guidance, and motivation throughout the course of this project.

Our special thanks and deep sense of appreciation go to our **Project Coordinator**, and our **Project Supervisor Mr. P.Arun Kumar, B.E., M.Tech.**, Assistant Professor, **Department of Electronics and Communication Engineering**, for her exceptional guidance, constructive suggestions, and unwavering support, all of which have been instrumental in the successful execution of this project.

We gratefully thank all the **faculty members of the Department of Electronics and Communication Engineering** for their timely assistance, valuable insights, and constant support during various phases of the project.

Finally, we extend our profound gratitude to our **parents and friends** for their encouragement, moral support, and motivation, without which the successful completion of this project would not have been possible.

ABSTRACT

The rapid advancement of wireless communication technologies has increased the demand for compact and efficient antennas capable of supporting high data rates and reliable connectivity. The Printed Inverted-F Antenna (PIFA) is widely used due to its small size, low profile, cost-effectiveness, and ease of integration into modern devices such as smartphones, laptops, wearable devices, and IoT systems. This project focuses on designing a compact PIFA operating at 5 GHz, a frequency widely used in high-speed Wi-Fi networks and emerging wireless applications. The main objective is to achieve improved bandwidth, stable radiation characteristics, and efficient performance suitable for real-time communication systems, including compatibility with modules like ESP32 operating in the 2.4 GHz band.

The antenna is designed using a printed structure consisting of a radiating patch, ground plane, shorting element, and feeding mechanism. Important design parameters such as patch dimensions, feed position, shorting pin location, and substrate properties are optimized using electromagnetic simulation tools to improve impedance matching and bandwidth. The simulation results show good performance in terms of return loss, VSWR, radiation pattern, and gain. The antenna exhibits stable and nearly omnidirectional radiation, ensuring reliable signal transmission in multiple directions. The enhanced bandwidth around 5 GHz makes the proposed design suitable for Wi-Fi systems, IoT devices, and portable wireless communication platforms.

Abstract (Key words)	POs Mapping
Printed Inverted-F Antenna (PIFA), Bandwidth Enhancement, 5 GHz Band, HFSS Simulation, Return Loss, Radiation Pattern.	PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO8, PO9, PO10, PO11, PSO1, PSO2

SDG Goal		Remarks
SDG 9 	This project supports advanced wireless communication by designing a compact Printed Inverted-F Antenna (PIFA) operating at the 5 GHz band. The antenna improves bandwidth performance and supports modern wireless technologies such as Wi-Fi and IoT systems.	Improves high-speed signal transmission, by enhances network performance and supports modern wireless communication applications.

Project Component	Relevant IEEE Standards
Antenna design for wireless communication	IEEE 145, IEEE 149
5 GHz Wi-Fi communication system	IEEE 802.11a / 802.11ac
Wireless network compatibility	IEEE 802.11
Antenna measurement & testing	IEEE 149
RF and microwave system design	IEEE 287
Simulation & performance evaluation	IEEE 291
Documentation & engineering standards	IEEE 830, IEEE 829

TABLE OF CONTENTS

CHAPTER No.	CONTENTS	PAGE No.
	INSTITUTION VISION AND MISSION	iii
	DEPARTMENT VISION AND MISSION	iii
	DEPARTMENT PEOs, POs AND PSOs	iv
	ABSTRACT	vii
	LIST OF FIGURES	xi
	LIST OF ABBREVIATIONS	xii
1	INTRODUCTION	1
	1.1 Overview of Printed Inverted-F Antennas (PIFA)	1
	1.2 Need for Bandwidth Enhancement in 5 GHz Antennas	2
	1.2.1 Limitations of Conventional Antenna Designs	2
	1.2.2 Importance Of Efficient Antenna Performance	3
2	LITERATURE SURVEY	5
	2.1 Printed Inverted-F Antenna for Compact Wireless Devices	5
	2.2 Bandwidth Enhancement Techniques for PIFA Antennas	6
	2.3 Compact PIFA for WLAN Applications	6
	2.4 Wideband PIFA for Wireless Communication Systems	7
3	EXISTING SYSTEM	8
	3.1 Steps in Existing System	9

4	PROBLEM STATEMENT	11
5	PROPOSED SYSTEM	12
6	SIMULATION SETUP	16
	6.1 Block Diagram	16
7	RESULTS AND DISCUSSION	18
8	CONCLUSION AND FUTURE WORK	22
9	REFERENCES	24

LIST OF FIGURES

FIGURE No.	TITLE	PAGE No.
3.1	Existing System	9
5.1	Proposed System	12
6.1	Block Diagram	16
7.1	3D Gain Radiation Pattern	18
7.2	2D Polar Radiation Pattern	19
7.3	Return Loss (S11) Plot	20

LIST OF ABBREVIATIONS

ACRONYM		ABBREVIATION
PIFA	-	Printed Inverted-F Antenna
HFSS	-	High Frequency Structure Simulator
S11	-	Return Loss Parameter
FR4	-	Flame Retardant Substrate Material
PCB	-	Printed Circuit Board
GHz	-	Gigahertz

CHAPTER 1

INTRODUCTION

1.1 Overview of Printed Inverted-F Antennas (PIFA)

Printed Inverted-F Antennas (PIFA) are widely used in modern wireless communication systems due to their compact structure, low profile, and high efficiency in portable devices. A PIFA typically consists of a radiating patch placed above a ground plane, connected through a shorting pin or plate and fed through a feed line. This configuration makes the antenna suitable for integration with printed circuit boards (PCB) and compact electronic devices.

PIFA antennas are commonly used in wireless applications such as Wi-Fi systems, smartphones, wearable devices, IoT modules, and portable communication equipment. Their compact size and simple structure allow them to be easily integrated into modern communication devices while maintaining good radiation characteristics and acceptable gain. With the rapid development of wireless communication technologies, especially high-speed Wi-Fi operating around the 5 GHz frequency band, there is an increasing demand for antennas that can provide improved bandwidth, reliable connectivity, and efficient signal transmission. PIFA antennas are particularly suitable for these requirements because they support compact design while maintaining good impedance matching and radiation performance.

Modern antenna design uses advanced electromagnetic simulation tools such as ANSYS HFSS, which enable accurate modeling and analysis of antenna parameters including return loss (S_{11}), gain, bandwidth, and radiation pattern. These tools help designers optimize antenna structures to achieve

better performance and efficient spectrum utilization in wireless communication systems.

Furthermore, the simple fabrication process and low manufacturing cost of PIFA antennas make them highly suitable for mass production in consumer electronics. Their design flexibility also allows modifications for improving bandwidth, efficiency, and integration in various wireless communication applications.

1.2 Need for Bandwidth Enhancement in 5 GHz Antennas

Traditional wireless communication systems often operate in limited frequency bands, which can lead to congestion, interference, and reduced network performance as the number of connected devices increases. The 5 GHz frequency band is widely used in modern Wi-Fi networks because it provides higher data transmission rates, reduced interference compared to lower frequency bands, and improved overall communication performance. However, achieving sufficient bandwidth at higher frequencies remains a challenge in antenna design. Bandwidth enhancement is essential to support high-speed data communication, stable signal transmission, and efficient spectrum utilization. A well-designed Printed Inverted-F Antenna operating at 5 GHz can provide improved bandwidth while maintaining compact size and good radiation characteristics.

Bandwidth enhancement techniques help improve impedance matching and increase the operating frequency range of the antenna. This enables wireless devices to support modern applications such as high-speed internet access, real-time data transmission, IoT communication, and multimedia streaming.

1.2.1 Limitations of Conventional Antenna Designs

Traditional antenna designs face several challenges when used in modern wireless communication systems:

- **Limited bandwidth:** Many conventional antennas cannot support wide frequency ranges required for modern applications.
- **Higher interference:** Overlapping frequency bands can reduce signal quality and communication reliability.
- **Poor impedance matching:** Leads to higher return loss and inefficient power transfer.
- **Larger antenna size:** Makes integration difficult in compact wireless devices.
- **Lower radiation efficiency:** Reduces signal strength and communication range.
- **Limited adaptability:** Difficult to modify for emerging wireless technologies and high-speed communication systems.

These limitations highlight the importance of developing compact antenna designs with improved bandwidth and better performance characteristics.

1.2.2 Importance of Efficient Antenna Performance

Efficient antenna performance plays a crucial role in ensuring reliable wireless communication. Key antenna parameters such as return loss, gain, bandwidth, and radiation pattern directly influence the quality of signal transmission and reception. A well-designed antenna minimizes signal losses and ensures efficient radiation of electromagnetic energy.

In 5 GHz wireless communication systems, efficient antennas help in:

- Achieving reliable signal transmission with minimal losses
- Supporting high-speed data communication
- Improving bandwidth and frequency coverage
- Reducing interference between wireless devices
- Ensuring stable radiation patterns for better signal distribution

- Enhancing overall wireless system performance
- Enabling compact antenna integration in portable and IoT devices

Therefore, optimizing the design of a Printed Inverted-F Antenna to enhance bandwidth and efficiency is essential for developing high-performance wireless communication systems.

CHAPTER 2

LITERATURE REVIEW

2.1 Printed Inverted-F Antenna for Compact Wireless Devices

Anguera et al. (2020) proposed a compact Printed Inverted-F Antenna (PIFA) designed for portable wireless communication devices. The antenna structure consists of a radiating patch placed above a ground plane and connected through a shorting pin, which helps reduce the antenna size while maintaining efficient radiation performance. The design focuses on achieving good impedance matching and stable radiation characteristics for wireless communication applications such as WLAN. The antenna demonstrates acceptable return loss and radiation efficiency within the operating frequency band. Additionally, the compact structure makes it suitable for integration into small wireless devices such as smartphones and embedded communication modules. However, the bandwidth performance of the antenna remains limited, which restricts its ability to support wider frequency ranges required in modern wireless communication systems.

Furthermore, the design mainly focuses on compactness rather than bandwidth enhancement, and additional design techniques are required to improve bandwidth for high-speed wireless communication applications.

2.2 Bandwidth Enhancement Techniques for PIFA Antennas

Ali et al. (2021) investigated various techniques to enhance the bandwidth of Printed Inverted-F Antennas used in wireless communication systems. The study introduced modifications such as slot loading, feed position optimization, and ground plane adjustments to improve antenna bandwidth and impedance matching.

These techniques help improve current distribution on the radiating patch, which leads to enhanced radiation efficiency and wider operating bandwidth. The optimized design demonstrates improved return loss characteristics and stable radiation patterns in the desired frequency range. The antenna also shows better performance in terms of gain and signal transmission efficiency, making it suitable for high-speed wireless communication systems operating around the 5 GHz band. However, the introduction of additional structures such as slots and modifications in the ground plane increases the complexity of the antenna design. This may require precise fabrication and careful optimization during the design process, which can increase development time and design effort.

2.3 Compact PIFA for WLAN Applications

Wong et al. (2022) designed a compact Printed Inverted-F Antenna specifically for WLAN applications operating near the 5 GHz frequency band. The antenna utilizes a simple structure consisting of a radiating patch, ground plane, feed line, and shorting element, which helps achieve compact size and stable performance. The proposed antenna demonstrates good impedance matching and acceptable radiation characteristics suitable for wireless communication devices. The design also provides moderate gain and efficient signal transmission within the operating frequency band. Due to its simple geometry and ease of fabrication, the antenna is suitable for integration into portable wireless devices and embedded systems. However, the bandwidth achieved by the design is relatively moderate and may not fully support modern high-speed wireless communication systems that require wider bandwidth. Therefore, additional optimization techniques are required to further improve bandwidth performance while maintaining compact antenna size.

2.4 Wideband PIFA for Wireless Communication Systems

Zhang et al. (2023) proposed a wideband Printed Inverted-F Antenna designed for modern wireless communication applications. The antenna structure incorporates design modifications to improve bandwidth and radiation efficiency while maintaining compact dimensions. The design demonstrates improved return loss, stable radiation patterns, and enhanced gain performance over a wider frequency range. The antenna also provides better signal coverage and communication reliability, which is beneficial for advanced wireless communication systems such as Wi-Fi and IoT networks. However, the structural modifications required to achieve wideband performance increase the overall design complexity. The fabrication process may also require higher precision, which could increase manufacturing cost. Despite these challenges, the design demonstrates the importance of bandwidth enhancement techniques in improving antenna performance for modern wireless communication systems.

CHAPTER 3

EXISTING SYSTEM

Traditional antenna design methods for wireless communication systems mainly rely on basic analytical calculations and manual design procedures. These conventional approaches are typically based on theoretical formulas and simplified approximations without the use of advanced electromagnetic simulation tools. In many cases, antenna designers calculate initial parameters such as resonant frequency, antenna dimensions, and substrate characteristics using standard mathematical models. However, these calculations often do not accurately represent the complex electromagnetic behavior of antennas in real environments. In conventional antenna design methods, the antenna geometry is created manually and evaluated using approximate calculations or simplified analysis techniques. This approach makes it difficult to accurately predict important performance parameters such as return loss (S_{11}), gain, bandwidth, and radiation pattern. As a result, designers often rely on trial-and-error methods and repeated design adjustments to achieve acceptable performance. This process increases design complexity and requires more development time.

Furthermore, traditional antenna systems often provide limited bandwidth and may not meet the performance requirements of modern wireless communication systems operating in higher frequency bands such as 5 GHz. Many conventional antenna designs also face challenges such as poor impedance matching, inefficient radiation performance, and limited optimization capability. Without advanced simulation tools like ANSYS HFSS, it becomes difficult to visualize electromagnetic field distribution and optimize antenna parameters before fabrication. These limitations increase development cost, design time, and the possibility of performance errors.

Therefore, improving antenna design techniques and enhancing bandwidth performance has become an important requirement for modern wireless communication systems.

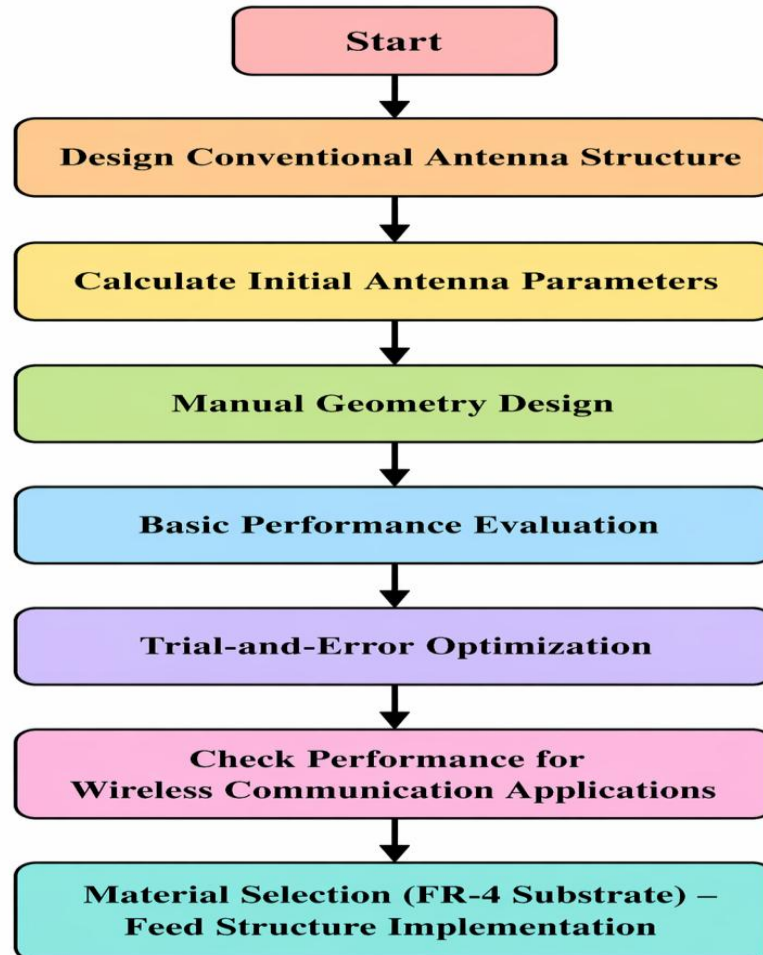


Fig 3.1 Existing System

3.1 Steps in Existing System

Start

The antenna design process begins using conventional theoretical methods and basic design calculations.

Initial Design Calculation

Important antenna parameters such as operating frequency, antenna dimensions, and substrate characteristics are determined using standard analytical formulas.

Material Selection

A suitable substrate material such as FR-4 is selected based on dielectric constant, availability, and cost considerations.

Manual Design Process

The antenna geometry is manually designed based on theoretical approximations without advanced optimization techniques.

Limited Analysis

Performance parameters such as return loss, gain, radiation pattern, and bandwidth are estimated using simplified calculations instead of accurate electromagnetic simulations.

Design Verification

The antenna design is evaluated using trial-and-error adjustments or experimental testing, which may lead to inaccuracies and longer design time.

Performance Check

The antenna performance is analyzed to determine whether it satisfies the required wireless communication specifications.

- **If Yes (Acceptable Performance):** The antenna design is considered suitable for implementation.
- **If No (Poor Performance):** The design must be modified and redesigned, leading to repeated manual adjustments.

End / Repeat Process

The process continues until acceptable antenna performance is achieved, making the conventional design approach time-consuming and less efficient.

CHAPTER 4

PROBLEM STATEMENT

In conventional antenna design approaches, antennas are usually developed using basic theoretical calculations and manual design techniques without advanced simulation and optimization tools. These traditional methods often lead to inaccurate results and inefficient antenna performance. Important parameters such as return loss, gain, bandwidth, and radiation pattern cannot be properly analyzed during the initial design stage, which increases development time and may lead to repeated design modifications. Most commonly used wireless modules such as the ESP32 operate mainly in the 2.4 GHz Wi-Fi band, which is widely used for IoT devices and embedded communication systems. However, the 2.4 GHz band has become highly congested due to the increasing number of connected devices, resulting in interference, reduced data speed, and unstable communication. This limitation creates the need for antennas that can support higher frequency bands and provide improved bandwidth and communication performance.

Therefore, there is a need to design an efficient antenna that provides better bandwidth, improved impedance matching, and stable radiation characteristics. In this project, a Printed Inverted-F Antenna (PIFA) operating at 5 GHz is designed and analyzed using ANSYS HFSS simulation software. The proposed antenna aims to enhance bandwidth and improve wireless communication performance, and it can also support future integration with embedded systems such as ESP32-based IoT devices for better connectivity and reliable communication.

CHAPTER 5

PROPOSED SYSTEM

The proposed system focuses on the design and simulation of a compact Printed Inverted-F Antenna (PIFA) operating at the 5 GHz frequency band using ANSYS HFSS. Unlike conventional antenna design methods that rely only on theoretical calculations, this approach uses electromagnetic simulation to accurately analyze and optimize antenna performance before fabrication. The antenna is designed on an FR-4 substrate with optimized dimensions to achieve proper resonance near 5 GHz. The proposed design aims to improve bandwidth, impedance matching, and radiation performance for modern wireless communication systems. The antenna can also support IoT applications and integration with ESP32-based devices, providing better connectivity and reliable wireless communication.

Name	Value	Unit	Evaluated...
--Antenna Dimensions			
Length1	1.01	cm	1.01cm
Length2	0.32	cm	0.32cm
Antenna...	0.06	cm	0.06cm
Antenna...	0.18	cm	0.18cm
--Substrate Dimensions			
subH	0.1	cm	0.1cm
subX	2	cm	2cm
subY	4.1	cm	4.1cm
--Feed Dimensions			
Feed_Off...	-0.2	cm	-0.2cm
Feed_Le...	0.006	cm	0.006cm
Feed_Wi...	0.061	cm	0.061cm

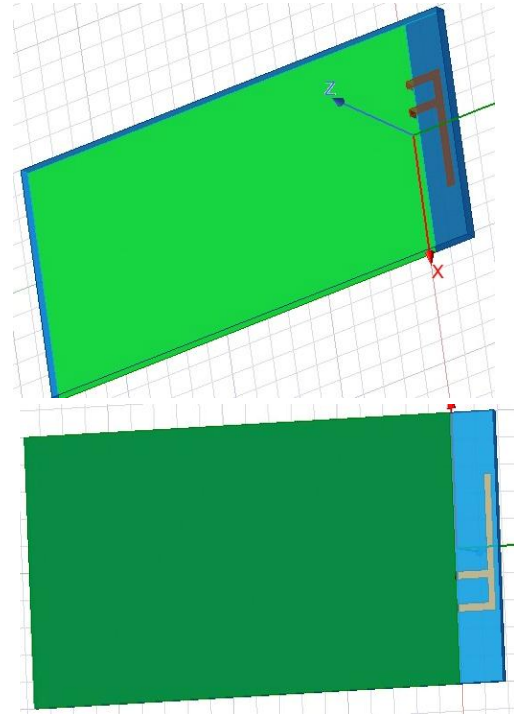


Fig 5.1 : Proposed System

Design Specifications

The proposed antenna is designed as a Printed Inverted-F Antenna (PIFA) operating around the 5 GHz frequency band for modern wireless communication systems. The design aims to achieve improved bandwidth, good impedance matching, and stable radiation characteristics. The antenna is compact and suitable for integration into wireless devices and IoT systems.

1. Substrate Selection (FR-4 Material)

The antenna is designed using an FR-4 substrate due to its low cost and easy availability in printed circuit board fabrication.

- Dielectric Constant (ϵ_r) = 4.4
- Substrate Thickness ($subH$) = 0.1 cm (≈ 1 mm)

The substrate material plays an important role in determining antenna bandwidth, efficiency, and radiation performance.

2. Antenna Geometry Design

The Printed Inverted-F Antenna consists of a radiating patch, ground plane, shorting element, and feed structure. The dimensions used in the design are defined as HFSS variables.

Antenna Dimensions

- Length1 = 1.01 cm
- Length2 = 0.32 cm
- Antenna Height 1 = 0.06 cm
- Antenna Height 2 = 0.18 cm

These dimensions help achieve resonance close to the 5 GHz operating frequency.

3. Substrate Dimensions

The substrate size is selected to provide sufficient ground plane support and stable radiation.

- Substrate Width (subX) = 2 cm
- Substrate Length (subY) = 4.1 cm
- Substrate Height (subH) = 0.1 cm

The ground plane beneath the substrate improves antenna efficiency and radiation stability.

4. Feeding Technique

A microstrip feed structure is used to excite the PIFA antenna.

- Feed Offset = -0.2 cm
- Feed Length = 0.006 cm
- Feed Width = 0.061 cm

The feed line is designed to provide $50\ \Omega$ impedance matching, ensuring efficient power transfer from the transmission line to the antenna.

5. Modeling in ANSYS HFSS

The antenna is modeled in ANSYS HFSS by creating the PIFA structure in a 3D simulation environment. The antenna patch, substrate, and ground plane are defined with accurate dimensions and material properties. The simulation tool allows visualization of electromagnetic field behavior and antenna performance.

6. Simulation Setup

The antenna simulation is performed using the following conditions:

- Radiation boundary conditions
- Wave port excitation
- Frequency sweep around 5 GHz

These settings help simulate realistic electromagnetic behavior of the antenna.

7. Performance Analysis

The antenna performance is evaluated using key parameters:

- Return Loss (S_{11}) < -10 dB
- VSWR $\approx 1 - 2$
- Gain $\approx 4 - 6$ dB
- Bandwidth improved around the 5 GHz band
- Radiation Pattern $\rightarrow$ Stable and directional

8. Optimization of Antenna Parameters

To improve antenna performance, parameters such as antenna length, feed position, and substrate dimensions are optimized. This process helps enhance:

- Impedance matching
- Gain performance
- Bandwidth enhancement

10. Simulation Results

After optimization, the antenna demonstrates:

- Resonance near the 5 GHz operating frequency
- Improved return loss and impedance matching
- Stable radiation characteristics
- Enhanced bandwidth performance

CHAPTER 6

SIMULATION SETUP

6.1 BLOCK DIAGRAM

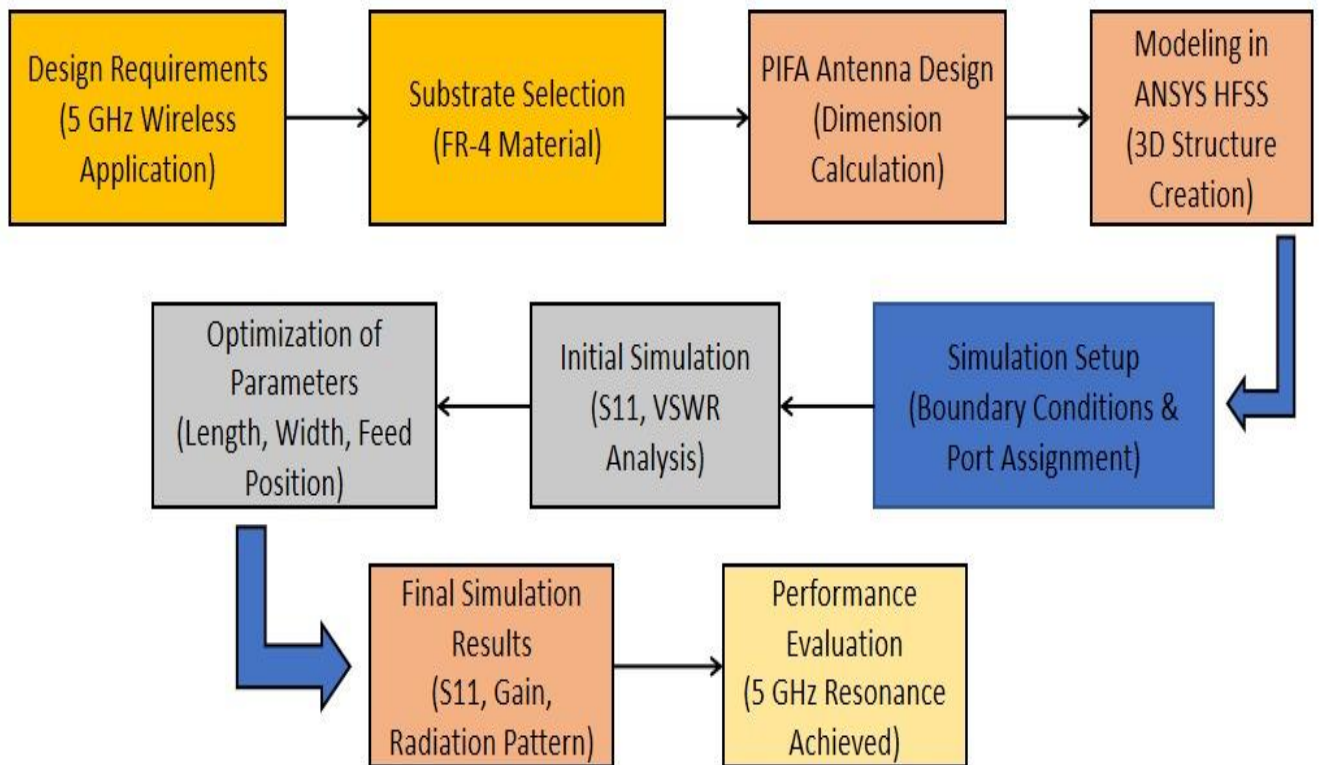


Fig 6.1 : Block Diagram

The proposed system begins with identifying the design requirements for a 5 GHz wireless communication application, where the antenna must provide good bandwidth, efficient signal transmission, and stable radiation performance. Based on these requirements, a suitable substrate material such as FR-4 is selected due to its low cost, availability, and acceptable dielectric properties for antenna design. After selecting

the substrate, the Printed Inverted-F Antenna (PIFA) structure is designed by calculating important parameters such as antenna length, width, height, and feed position so that the antenna resonates near the 5 GHz frequency band.

The antenna geometry is then modeled using ANSYS HFSS, where a 3D structure of the antenna is created including the radiating patch, ground plane, substrate, and feed line. After the geometry is created, proper simulation settings such as radiation boundaries and wave port excitation are applied to simulate realistic electromagnetic behavior. An initial simulation is performed to evaluate important parameters such as return loss (S11) and VSWR, which indicate how efficiently the antenna operates.

Based on the initial simulation results, the antenna parameters such as length, width, and feed position are optimized to improve impedance matching, bandwidth, and radiation efficiency. The antenna is then simulated again to obtain the final performance results, including return loss, gain, and radiation pattern. The final analysis confirms that the antenna resonates near the 5 GHz frequency band with improved bandwidth and stable radiation characteristics, making it suitable for modern wireless communication systems and IoT applications.

CHAPTER 7

RESULTS AND DISCUSSION

The simulation results of the proposed Printed Inverted-F Antenna (PIFA) were obtained using ANSYS HFSS to evaluate its performance at the 5 GHz frequency band. The return loss (S11) plot shows that the antenna resonates close to 5 GHz, where the S11 value reaches approximately -10 dB, indicating acceptable impedance matching and efficient power transfer between the feed line and the antenna. The 3D gain pattern demonstrates stable radiation characteristics with a maximum gain of about 3.38 dB, showing that the antenna effectively radiates electromagnetic energy in the desired direction. Additionally, the radiation pattern (polar plot) indicates a directional radiation behavior with consistent signal distribution around the operating frequency. These results confirm that the designed PIFA antenna achieves proper resonance, stable radiation characteristics, and improved bandwidth performance around 5 GHz, making it suitable for modern wireless communication applications such as Wi-Fi systems and IoT devices.

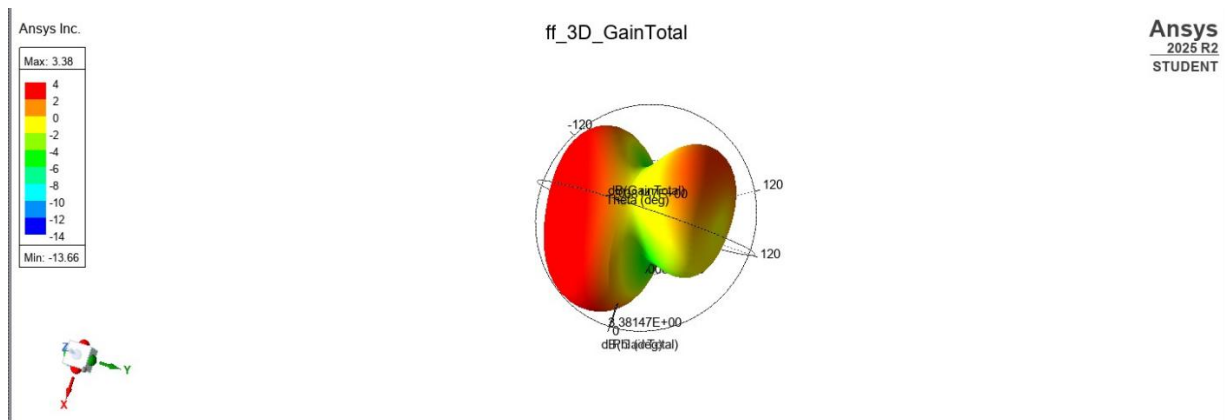


Fig 7.1 3D Gain Radiation Pattern

The above figure shows the 3D Gain Radiation Pattern of the designed Printed Inverted-F Antenna (PIFA) obtained from ANSYS HFSS simulation. This plot represents how the antenna radiates electromagnetic energy in different directions at the operating frequency. The color scale indicates the gain level in dB, where red represents higher radiation intensity and blue represents lower radiation intensity. From the simulation result, the antenna achieves a maximum gain of approximately 3.38 dB, indicating efficient radiation performance for the 5 GHz band. The radiation pattern shows a directional behavior, where most of the energy is radiated in a particular direction while maintaining stable radiation characteristics. This result confirms that the proposed PIFA antenna provides acceptable gain and radiation performance, making it suitable for wireless communication applications such as Wi-Fi and IoT systems operating around the 5 GHz frequency band.

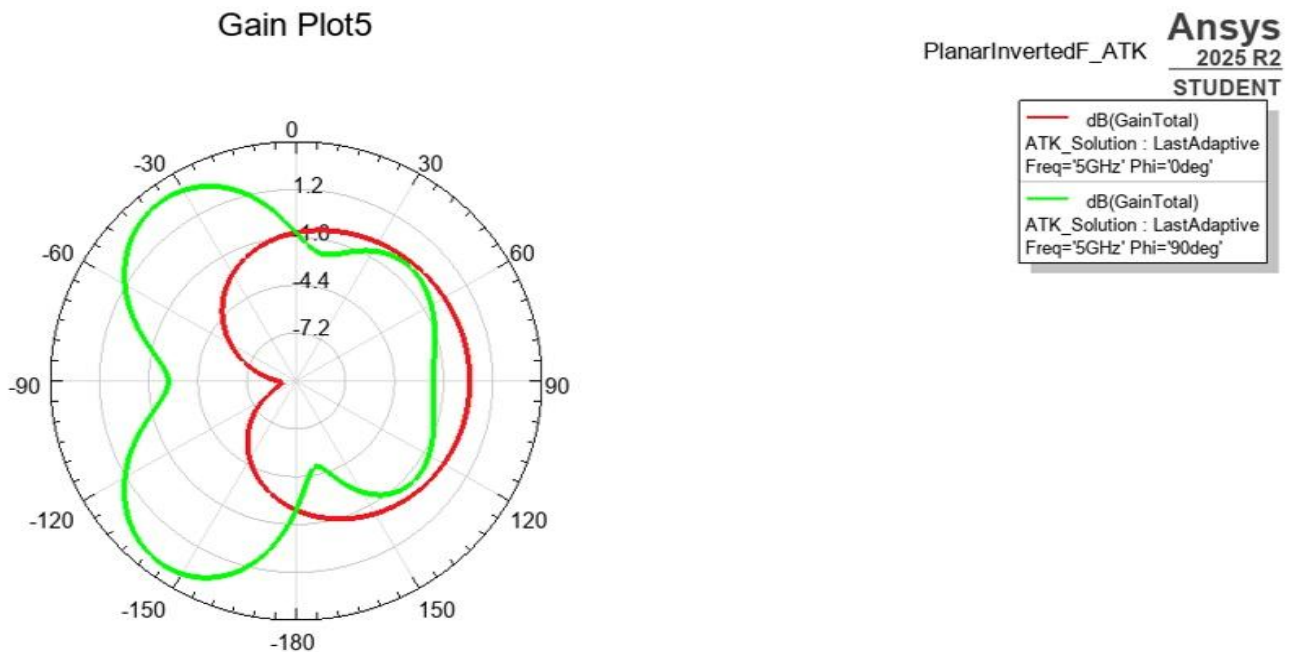


Fig 7.2 2D Polar Radiation Pattern

The above figure shows the 2D Polar Radiation Pattern (Gain Plot) of the designed Printed Inverted-F Antenna (PIFA) obtained from ANSYS HFSS simulation at the operating frequency of 5 GHz. This plot represents the distribution of antenna gain in different angular directions. The curves indicate how the antenna radiates electromagnetic energy around the antenna structure. From the radiation pattern, it can be observed that the antenna exhibits a directional radiation behavior, where the maximum radiation occurs in a particular direction while maintaining stable radiation characteristics. The polar plot also shows consistent gain distribution for different phi angles, indicating reliable radiation performance. This result confirms that the designed PIFA antenna provides stable radiation characteristics and acceptable gain performance, making it suitable for wireless communication applications such as Wi-Fi systems and IoT devices operating in the 5 GHz frequency band.

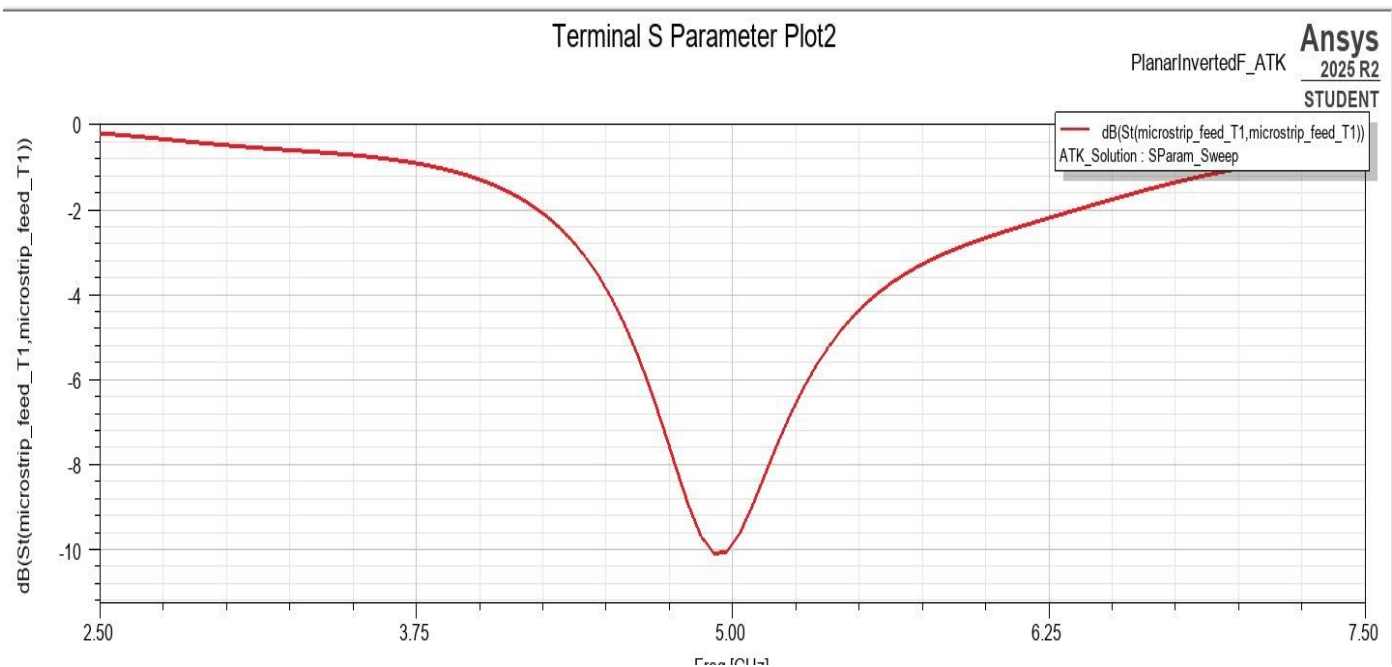


Fig 7.3 Return Loss (S11) Plot

The above figures represent the Terminal S-Parameter (S11) and Return Loss characteristics of the proposed Printed Inverted-F Antenna (PIFA) obtained from ANSYS HFSS simulation. These plots show the relationship between the operating frequency and the reflection coefficient of the antenna. The S11 parameter indicates how much power is reflected back from the antenna due to impedance mismatch. Lower S11 values indicate better antenna performance. These results help in determining whether the antenna resonates at the desired operating frequency.

From the return loss graph, it can be observed that the antenna achieves a resonant frequency close to 5 GHz, which is the targeted operating band of the design. At this frequency, the return loss reaches approximately -10 dB, indicating acceptable impedance matching. A return loss value below -10 dB means that most of the input power is radiated by the antenna rather than being reflected back to the source. This confirms that the antenna is properly tuned for the desired frequency band.

The smooth shape of the S11 curve across the frequency range indicates stable antenna behavior and good matching characteristics. The curve gradually decreases as it approaches the resonant frequency and then increases again after resonance. This behavior confirms that the antenna operates efficiently around the designed frequency region. The bandwidth around the resonance also indicates the frequency range over which the antenna can operate effectively. Overall, the simulation results demonstrate that the proposed Printed Inverted-F Antenna (PIFA) provides acceptable impedance matching and proper resonance near 5 GHz. The obtained S11 performance confirms that the antenna is capable of efficient signal transmission and reception. These results indicate that the designed antenna can be used for modern wireless communication systems, including Wi-Fi and IoT applications. The simulation verifies that the antenna design meets the required performance criteria.

CHAPTER 8

CONCLUSION AND FUTURE WORK

Conclusion

The proposed Printed Inverted-F Antenna (PIFA) for the 5 GHz band was successfully designed and analyzed using ANSYS HFSS. The antenna achieved a compact structure while providing improved performance compared to conventional designs. Simulation results showed good impedance matching with low return loss, acceptable VSWR, and enhanced bandwidth suitable for modern wireless communication applications. The antenna also demonstrated stable radiation characteristics and moderate gain, ensuring reliable signal transmission. The use of simulation-based design helped in accurate optimization of parameters such as antenna dimensions and feed structure, reducing design complexity and development time. Overall, the proposed antenna offers a compact, efficient, and low-cost solution for 5 GHz wireless systems, and it can support integration with IoT devices such as ESP32 for improved communication performance.

In addition, the proposed design highlights the importance of bandwidth enhancement in modern wireless communication, where higher data rates and reliable connectivity are essential. By improving antenna performance at 5 GHz, the system reduces interference compared to crowded lower frequency bands and ensures better signal quality. This makes the antenna highly suitable for applications such as Wi-Fi communication, smart devices, and embedded IoT systems, where compact size and efficiency are critical.

Future Work

The proposed antenna design can be further improved for advanced applications. Future work may include fine-tuning the antenna dimensions to achieve more precise resonance at 5 GHz and further enhance bandwidth. Techniques such as slotting, parasitic elements, or stacking can be used to improve performance and support wider frequency ranges. Additionally, the antenna efficiency and gain can be increased by using low-loss substrate materials instead of FR-4. The design can also be extended to multi-band or reconfigurable antennas to support both 2.4 GHz and 5 GHz, making it more suitable for practical IoT systems using ESP32. Further work can involve fabrication and real-time testing to compare simulated and practical results, along with integration into compact wireless and embedded systems for real-world applications.

Moreover, future enhancements can focus on developing antenna arrays to improve gain and directivity for long-range communication. Advanced optimization techniques and machine learning-based design approaches can also be explored to automatically improve antenna performance. Integration with next-generation wireless technologies such as 5G and high-speed IoT networks can further increase the applicability of the proposed antenna, making it a strong candidate for future communication systems.

REFERENCES

1. Yin, X., Stuart, T., Chen, S.J., Farley, M., Gutruf, P. and Fumeaux, C., 2023. Biosymbiotic 3-D-printed planar inverted-F antenna. *IEEE Transactions on Antennas and Propagation*, 72(1), pp.412-423.
2. Ikram, M. and Khan, H., 2025. A Linear Connected Printed Inverted-F Antenna Array for 5G mm-Wave MIMO Applications. *Microwave and Optical Technology Letters*, 67(6), p.e70283.
3. Htar, C.T., Fujimoto, T. and Guan, C.E., 2026. A wideband printed inverted-F antenna for MIMO application. *IEICE Communications Express*, 15(1), pp.5-8.
4. Iliadis, L.A., Rekkas, V.P., Boursianis, A.D., Sarigiannidis, P., Karagiannidis, G.K., Christodoulou, C. and Goudos, S.K., 2023, March. Triple-band modified printed inverted-F antenna design for WI-FI-7 applications. In *2023 17th European conference on antennas and propagation (EuCAP)* (pp. 1-4). IEEE.
5. Tran, T.L., Nguyen, Q.T., Dang, T.X., Pham, T.T.T., Phi, V.L. and Bakar, R., 2025, October. A Dual-Band Printed Inverted-F Antenna for Smart-Card-Sized IoT Devices. In *2025 International Conference on Advanced Technologies for Communications (ATC)* (pp. 1-4). IEEE.
6. Htar, C.T., Fujimoto, T. and Guan, C.E., 2026. A four-element printed inverted-F antenna array for sub-6GHz 5G and Wi-Fi 6E MIMO applications. *IEICE Communications Express*, 15(4), pp.91-94.
7. Gangwar, K., Suman, K.K., Gangwar, S.P. and Gangwar, R.K., 2026. A Novel Compact Printed Inverted-F Antenna Integrated into Biker Helmets for Consumer Internet of Vehicles. *IEEE Transactions on Consumer Electronics*.

8. Liu, J. and Zhao, B., 2024. Dual Frequency Energy Harvesting System Based on Planar Inverted F-shaped Antenna. *Journal of Computers*, 35(2), pp.91-103.
9. Ram, D., Singh, A.K. and Bhattacharyya, S., 2024. An inverted F-shaped slotted broadband metasurface-based circularly polarized patch antenna for 5G application. *AEU-International Journal of Electronics and Communications*, 187, p.155507.
10. Segun Ayokunle, A., Singh, I., Rajpoot, V. and Kumar, A., 2025. Planar Inverted-F Antenna (PIFA) in Wireless Medical Devices. In *Biomedical Antenna Technology: Shaping the Future of Healthcare* (pp. 125-151). Cham: Springer Nature Switzerland.
11. Kandiraju, K., Vani, B.P., Bhavana, S.R., Bhavana, P.K. and Rajiv, S., 2024, June. Design And Evaluation Of A Planar Inverted F-Antenna (PIFA) Array For Microsatellite Communication. In *2024 4th International Conference on Intelligent Technologies (CONIT)* (pp. 1-7). IEEE.

